Linseed

Technology for Increasing Production

ICAR-Project Coordinating Unit (Linseed)

All India Coordinated Research Project on Linseed
Indian Institute of Pulse Research
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CONTENTS

INTRODUCTION	01
AREA OF CULTIVATION	01
SEASON AND SOIL	02
VARIETIES	02
CROPPING SYSTEM	02
Sole cropping	02
Intercropping	02
Paira or *Utera* cropping	03
Tips to obtain higher yield in *Utera* cultivation	04
AGRONOMIC MANAGEMENT	04
Land preparation	04
Fertilizer application	05
Sowing time	05
Spacing and seed rates	05
Seeding depth	05
Seed treatment	05
Irrigation	05
Weed control	05
PLANT PROTECTION	06
HARVESTING	06
STORAGE	06
TIPS TO OBTAIN HIGHER YIELD	09
FIBRE FROM DRY STALK OF LINSEED/ FLAX	09
Manual	10
Mechanical	10
MEDICINAL PROPERTIES	13
IMPACT OF IMPROVED TECHNOLOGIES (FLD's)	33

INTRODUCTION

Linseed (*Linum usitatissimum* L.) is an important oilseed crop grown for both seed and fibre. In South-west Asia including Turkistan, Afghanistan and India, it is primarily grown for oil. In Asia Minor, South Russia and America, it is traditionally cultivated for oil meant for edible as well as industrial purposes. Almost every part of its plant is commercially utilized either direct or after processing. It is next in importance to rapeseed-mustard grown during *rabi* season. Presently, it is grown on 2.94 lakh hectares contributing 1.54 lakh tons in production in the oilseed scenario of the country with a productivity level of 525 kg/ha (Annual report of linseed 2017-18). Linseed occupies an important position in India for its technical grade vegetable oil producing ability. Fibre yielding property from the linseed stalk is an added advantage of this crop.

On small scale, the seed and its oil are directly used for human consumption as flax seed breads, bagels and other baked and fried food stuffs by a small segment of population. It is industrial oil and 80 per cent of oil is used for paints, varnishes, a wide range of coating oils, linoleum, pad and printing inks, leather and soap industries. Linseed is highly nutritious. It is a good source of complete protein (all 8 essential amino acids), high order linolenic acid (an essential polyunsaturated Omega-3 fatty acid), complex carbohydrates, vitamins and minerals. Recent advances in medical research have found linseed as best herbal source of Omega-3 and Omega-6 fatty acids which have immense nutritional/medicinal effect on human body system. The oil cake is a good feed for milch cattle. Linseed stem yields good quality fibre having strength and durability. The fibre is lustrous and blends very well with wool, cotton and silk, *etc*. Strong twines, canvas, suiting, shirting and various indispensable products for defense purposes are manufactured from it. The woody matter and short fibres may be used as raw pulp for making paper of quality. The rough and strong linseed fibre can effectively be used for low-cost roofing tiles based on convertible polymers and for fibre-reinforced plastics based on unsaturated polyesters. The present productivity level of this crop (525 kg/ha) can be increased more than three times, if it is grown under the following recommended package of practices, reported from front line demonstration (Annexure-1).

AREAS OF CULTIVATION

The major linseed growing states are Madhya Pradesh, Chhattisgarh, Maharashtra, Uttar Pradesh, Odisha, Jharkhand, Bihar and Rajasthan. In

addition to the above, it is also grown in Karnataka, Nagaland, Assam, West Bengal, Jammu & Kashmir, Himachal Pradesh, Punjab and Meghalaya.

SEASON AND SOIL

Linseed is purely a *rabi* crop. It is often grown on marginal and sub-marginal rainfed soils. It can be grown successfully on different soils ranging from light soil to heavy one. If thrives well wet soil pH 5-7 but pH 6 is best. Drought and high temperature (>32°C) during and after the flowering stage reduce the yield. However, 20-30°C temperature is ideal.

VARIETIES

There are number of varieties available for cultivation in different cropping systems, situations and for varied agro-climatic conditions in linseed. A total of nine double purpose varieties were also developed which are having good seed and fibre yielding potentials. There is an urgent need to replace the old varieties from the farmers' field with new improved high yielding and disease resistant varieties. Details about the varieties suitable for different regions and situations are given in Table-1.

CROPPING SYSTEMS

Sole Cropping

Seed type of linseed can be grown either as sole crop or intercrop under rainfed and irrigated conditions with as well as double purpose type (seed & fibre producing) linseed can be grown under irrigated conditions. Double purpose linseed can compete well with other crops in terms of net monetary return, if it is grown as pure crop were recommended packages of practices Though, linseed yields 20 q/ha seed and 12 q/ha fibre in experimental fields on average basis, the national average is very low. There is a vast gap between the achievable yield and the national average as has been demonstrated through Frontline Demonstrations at farmer's field. In the light of its commercial, medicinal and nutritional importance, every effort should be made to use irrigation, fertilizers, plant protection measures, etc., to realize its optimum yield for seed and fibre both.

Intercropping

Intercropping of linseed particularly with chickpea, lentil, durum wheat, barley and safflower under rainfed condition and with wheat, potato, rapeseed & mustard needs to be popularized on a priority basis. This will provide stability to total production per unit land and may fetch higher net monetary returns. Intercropping linseed with chickpea may reduce wilt and pod-borer damage in chickpea. Growing linseed in intercropping system should be popularized in

place of mixed cropping. Some of the efficient intercrop combinations for various states are indicated as follows:

State	Situation	Intercropping system
Uttar Pradesh (Excluding Bundelkhand)	Rainfed	Linseed + Chickpea / Lentil (3 : 1 or 1 : 3) or (2:4)
	Irrigated	Linseed + Wheat (4 : 2), linseed + chickpea (2:4)
		Linseed + Mustard (5 : 1)
		Linseed + Potato (3 : 3)
Bundelkhand of U.P.	Rainfed	Linseed + Chickpea / Lentil (3 : 1 or 1 : 3) or (2:4)
		Linseed + Wheat (1 : 3)
Madhya Pradesh and Chhattisgarh	Rainfed	Linseed + Chickpea (3 : 1 or 4 : 4)
Bihar, Jharkhand & Assam	Rainfed	Linseed + Chickpea (4:2)
		Linseed + Mustard (5 : 1)
		Linseed + Lentil (3:3)
West Bengal	Irrigated	Linseed + Mustard (5 : 1)
		Linseed + Potato (3 : 3)
Maharashtra and Karnataka	Rainfed	Linseed + Chickpea (3 : 1 or 1 : 3) or (2:4)
		Linseed + Safflower (different row ratio)
Punjab and H.P.	Rainfed	Linseed + Chickpea (3 : 1 or 2:4)
	Irrigated	Linseed + Mustard (5 : 1)
		Linseed + Wheat (4 : 2)

*Varieties for Linseed & chickpea (2:4) intercropping system (Nagpur: PKVDIL - 260, Sagar : Padmini, Raichur : Indira Asli 32, Koh : Kof : Suyog/126 914, Shillogani : Shelechar/Serts/Shubbra, Tikamgarh : Padmini /TLS-67)

Paira or *Utera* cropping

This is one of the best practices for utilizing the residual moisture in rice fields where tillage is difficult. About 25% of the linseed area is under *utera* cropping. In this practice, linseed is broadcasted in the standing paddy fields at physiological maturity stage. Linseed is generally allowed to complete its life cycle under the moisture stress, with inadequate nutrients and plant

protection measures, resulting in poor yields.

Tips to obtain higher yield in *utera* cultivation

- RLC-143 (*Utera Alsi*) and RLC-153 (*Utera Alsi 2*) should be grown in Chhattisgarh, Madhya Pradesh and adjoining areas of Odisha whereas, Baner, Surabhi, LC-185 and Bhagsu in Himachal Pradesh.
- Heavy-textured soils with good water-retention capacity should be preferred.
- Sufficient farm yard manure (FYM) or green manure should be applied along with phosphatic fertilizers to paddy.
- A dose of 10 to 20 kg N/ha should be applied 2 or 3 days before linseed is sown.
- The seed rate should be 35-40 kg/ha in M.P., Chhattisgarh, Maharashtra, Odisha, and 50-60 kg/ha in Himachal Pradesh and Balaghat area.
- *Cuscuta* infestation, particularly in the Chhattisgarh, causes heavy losses. To ensure clean cultivation, *Cuscuta* seeds should be separated and discarded from the seed lot before sowing linseed.
- Sowing linseed when rice is at the dough stage or physiological maturity stage has proved to be the best. However, sowing may be done 10 to 12 days before rice is harvested, but should be completed by October.
- The weather should be clear at the time of sowing. Rains affect the plant stand adversely.
- Manual weeding should be done once or twice.
- Crack system of sowing is a new method which can be followed in areas where sufficient water is available. In this method, 5 cm deep cracks are allowed to develop in the field when the paddy crop is at the boot-leaf or panicle-initiation stage, and the field is watered thereafter. After keeping the water standing for 5 to 7 days, the normal practice of *utera* is followed. This method would give 60-100% more yield and has been found to have no adverse effect on paddy yields.
- Under *utera* system, linseed should be popularized as a mixed crop with greengram, blackgram and lathyrus, which are under cultivation in the system.

AGRONOMIC MANAGEMENT

Land preparation

The land should be ploughed 2 to 3 times followed by 2 to 3 harrowing for fine tilth. To conserve moisture, it is advisable to create soil mulch with the help of hoe after each good shower.

Fertilizer application

Linseed removes 60 kg N + 18.6 kg P_2O_5 + 54 kg K2O + 5.6 kg S + 31 kg Calcium + 13 kg Mg + 46 g Zn + 664 g Fe + 177 g Mn + 13 g Cu per tonne of produce and responds well to fertilizers. The recommended doses vary from state to state (Table-2). Under rainfed condition, all the fertilizers should be applied at the time of sowing whereas, under irrigated conditions, half dose of nitrogen and full dose of phosphorus and potash is applied as a basal dose. The remaining nitrogen is applied with the first irrigation *i.e.*, about 35 days after sowing. An application of 5t FYM/ha can save 25% of inorganic fertilizer. Seed inoculation with N bio fertilizer + PSB and 5 t FYM/ha can save 25% fertilizer in double crop system.

Sowing time

The sowing time varies widely (October to November) in different states, depending upon the availability of soil moisture and water for irrigation (Table-2). Early sowing helps the crop to escape attack of powdery mildew, rust, linseed bud fly in different regions.

Spacing and seed rates

The spacing and seed rates recommended for different states are given in Table-2.

Seeding depth

Depending upon the soil moisture, the seed should be placed 2-3 cm below the soil. Shallow sowing in well prepared land having sufficient moisture ensures good plant stand.

Seed treatment

Seed should be treated with Bavistin @ 1.5 g/kg seed or Thiram @ 3 g/kg seed or Topsin-M @ 2.5 g/kg or *Trichoderma viride or T. harzianum* 4 g/kg seed to protect the crop from seed-borne diseases and to some extent from soil-borne diseases also.

Irrigation

It is generally grown where annual precipitation ranges from 480-760 mm and it response well to irrigation. Yield can be doubled if two irrigations are given, the first at about 35 days after sowing and the second at about 65 days after sowing. However, three irrigations (at about 35, 55 and 75 days after sowing) have proved very effective in increasing the yield.

Weed control

Pre- emergence application of pendimethalin @ 1 kg/ha provide good control of most of the weeds, whereas post emergence application of clodinafop @ 60-80g/ha may manage weeds in the areas where isoproturon is not working for the control of *Phalaris minor*. Application of metasulfuron methyl @ 4g a.i./ha at 2-3 leaf stage of weed or post emergence for the effective control of *Medicago denticulta*.

PLANT PROTECTION

More than two dozen insect-pests have been reported damaging linseed crop right from seedling to harvest stage. Among these, bud fly (*D. lini*) is designated as the key pest throughout the linseed growing areas, while leaf miner, defoliators (semi-looper, lucerne caterpillar, Bihar hairy caterpillar) and gram pod borer are considered as occasional or sporadic pests. However, incidence of termite and cut worm may cause serious loss during seedling stage of the crop especially under rain fed situation. Extent of yield losses in linseed have been estimated ranging 7-26%, 19-97% and 33-98% by leaf miner, bud fly and gram pod borer, respectively. Suitable crop stages coinciding with favorable weather conditions trigger an epidemic of different insect-pests. Average temperature between 16-20ºC and relative humidity between 60-70% found suitable for best multiplication of linseed bud fly. Egg, larval, pupal and adult stages in one generation of bud fly complete in 2-5, 5-14, 5-11 and 1-3 days, respectively. One generation of this pest completes in 14-27 days and there are four overlapping generation in one cropping season. A workable management schedule for the insect-pests especially bud-fly is given in Table 3.

There are a number of diseases to which linseed or flax is exposed but Rust, Wilt, *Alternaria* blight and Powdery mildew are the major ones causing losses up to 80-100%, 80%, 27-60% and 60% respectively. Rust is a serious disease problem in Jammu and Kashmir, Himachal Pradesh, Punjab, Haryana and hilly regions of Uttar Pradesh; Wilt is serious in semiarid regions of Central India including Madhya Pradesh, Rajasthan, Uttar Pradesh and Bihar; *Alternaria* blight in Northern plains with high humidity *i.e.*, Uttar Pradesh, Jharkhand, Chhattisgarh and West Bengal; powdery mildew in Central and peninsular regions viz., Madhya Pradesh, Odisha, Maharashtra, Rajasthan and Karnataka. A phanerogamic plant parasite *Cuscuta* is a serious problem in Chhattisgarh and adjoining parts of Odisha and Maharashtra. The packages for management of diseases are illustrated in Table 4.

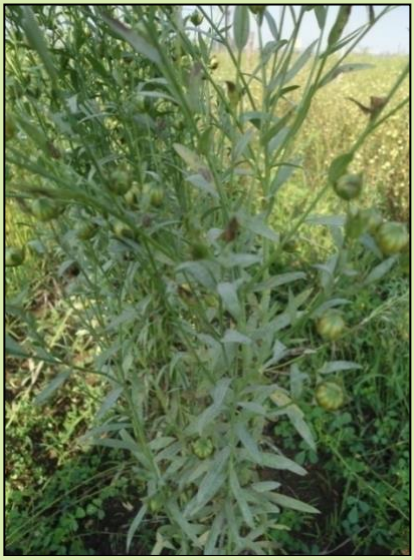
HARVESTING

The crop should be harvested when the leaves are dry, the capsules have turned brown and the seeds have become shiny.

STORAGE

Initial moisture and storage conditions are the most important factor which affects the stability of linseed in storage. A relative humidity of 70% with 8% moisture content in the seed is the best conditions for its storage.

Major diseases of Linseed



Powdery mildew

***Alternaria* blight**

Linseed wilt

Rust

Major Insect-pests of Linseed

Linseed Bud infested by Bud fly (L.)
And *Alternaria* blight

Semi Looper

Gram Pod Borer

Bihar Hairy Caterpilla

TIPS TO OBTAIN HIGHER YIELD

- Grow recommended resistant variety.
- Use quality seed of the recommended varieties.
- Treat the seed with fungicide, bactericide, *Trichoderma*, PSB, *Azotobactor etc.*, as recommended.
- At the time of sowing appropriate soil moisture should be ascertained in the field.
- Take up sowing at appropriate time.
- Keep the field weed free at least up to 40 days after sowing. Apply recommended doses of fertilizers at appropriate time.
- Adopt need based recommended plant protection measures against insect pests and diseases.
- Provide protective irrigation, if required.
- Harvest the crop at physiological maturity.

FIBRE FROM DRY STALK OF LINSEED / FLAX

The double purpose linseed varieties are suitable for seed as well as fibre production. A technical plant height (height from root to first branching) more than 50 cm. is categorized for double purpose varieties. The double purpose varieties are listed in Table-1. Fibre of linseed is one of the natural and ecofriendly, fibre having more strength, fineness and durability than many other fibres. It is soft, lustrous and flexible and has high water absorbency and blends very well with wool, cotton, silk, etc. It is being used in the preparation of strong twines, canvas, hosepipe, suiting-shirting and various indispensable products for aero-space and aeronautical and defense purposes. Woody core of stem and short fibre are used as raw pulp for making paper of quality. Fibre from linseed plant and different products of linseed fibre are presented in **Plate 3 and 4**. Know how about obtaining the fibres from linseed's stalks are illustrated in following steps.

1. Harvest the crop from ground level.
2. Make small bundles of the harvested crop, thresh out the seed and cut the stalk from first branching to separate the important unbranched stalk or cut the stalk from first branching and separate the seed portion and stalk to each other thereafter thresh out for seed.
3. Keep out the cut dry stalk separately for its retting.
4. The retting of stalks can be done by different methods viz; water retting, enzyme retting, chemical retting, dew retting, flash hydrolysis, gel retting, but this can be done easily at farmers home/farm involving following steps.
   (a) Place the stalk bundles in retting tank filled with water and leave it for 2-3 days. If stalks are floating put some weight on it so that stalks fully dip into the water.

(b) Remove the stalks from tank and wash them about 8-10 times with water thereafter allow it for sundry in open air.

5. Now the stalks are ready for scutching (separating the fibre from retted stalk). The scutching of fibre may be done with using either of following procedures.

Manual

1. Take small bundle of retted and dried stalk and beat it by hand mallet (mungri) on plane and hard surface. Owing to this the wooden part of the stalk will split out and resultant to this fibre can be obtained easily.
2. This type of suctching can be done at small scale right at farmer's home.

Mechanical (Fibre scutching machine)

1. Small bundle of retted and dried stalks placed on the feeder side of the machine and allow to run the machine (manually or power operated). The stalks get ruptured and come out from the delivery side.
2. Shake the ruptured stalk by hand resulting to this fibre, which will remain in hand and hard stick split out.
3. If the rupturing of the stalk is not completed in one time, repeat the process again to get the fibre completely free from the wooden part of the stalk.
4. Now, the power operated machines have been developed by which a large quantity of stalks can be scutched in a short time.

Hacking

The scutched fibres are combed in order to remove short fibres and extraneous matter. Hackling can be done by hand to remove short fibres, Trash and knock out shove from scutched fibre.

FLAX TO FIBRE

USES

Strong rope, blanket, carpet, matting, footmat, sleeping bed, newar, suiting & shirting, bedsheet, curtain, lace and other articles like paper, currency paper, cigarette paper, parchment paper, etc.

VALUE ADDED PRODUCTS

Plan Powder

Dry Chatni

Mukhwas

Biscuit

Cake

Fruit salad

Use of Linseed Stem (Cloth)

MEDICINAL PROPERTIES

The whole seed of linseed is rich in protein, fat, minerals, vitamins, etc. The major constituents of linseed are -

S. No.	Constituents	Proportion
1.	Moisture	6.5%
2.	Protein	20.3%
3.	Fat	37.1%
4.	Minerals	2.4%
5.	Carbohydrate	28.9%
6.	Energy	530 K Cal.
7.	Calcium	170 mg/100 g
8.	Iron	370 mg/100 g
9.	Carotene	2.7 mg/100 g
10.	Thiamine	0.23 mg/100 g
11.	Riboflavin	0.07 mg/100 g
12.	Niacin	1.0 mg/100 g

Linseed is a best source of omega-3 fatty acid and it is essential as it cannot be synthesized by the body, but must be supplemented directly from foods. This imparts in cholesterol lowering, cardiovascular benefits by affecting prostaglandins and leukotriens related to blood clotting and inflammatory disorder like rheumatoids arthritis. Seed works as bowel cleaning agent. Linseed is one of the richest sources of lignans; provide protection against certain form of cancer due to estrogenic and anti-estrogenic activity in the body. It has been reported that 30 g of milled seed provide adequate amount of a linolenic acid (Omega-3) and lignans. Linseed Tea which contains mucilage provides relief during cough and sneezing. The whole seed can be consumed directly after some heat treatment. It also works as mouth freshener can be use in the place of Pan Masala or Betel. The linseed powder can be consumed directly, mixing with flour for preparation of *chapati*, mixed with cooked pulse, etc. It is better to use fresh grind powder if not possible it should store in air tight bottle. The ground powder should be consumed in a week to avoid oxidation which can affect the quality of powder. Now days, people are aware about their health and they are consuming the raw seed, flax seed breads, bagels and other backed and fried food stuff due to it nutritional/medicinal properties. It is available in small packets of 50, 100, 200 g at big malls. The oil of linseed is being sold in capsules for supplement of omega-3 and omega-6.

Importance of linseed for health

Linseed is scientifically known as *Linum usitatissimum* L. and in Latin *usitatissimum* means 'most useful'. In modern world peoples are crazy for linseed seed due to its excellence and multifarious health benefits. It is highly nutritious, unique and emerging among oilseeds for its technical grade vegetable oil producing ability and containing 35-45% oil. It is a premium food delivering high performance and is one of the richest sources of omega 3 fatty acids known to man. Whole linseed, oil, lignin precursors and its mucilage have many potential uses in the prevention or treatment of various diseases. Linseed oil contains three times as much omega 3 fatty acid than omega 6 fatty acid. Linseed is nature's miraculous plant containing omega 3 fatty acids that are abundant in fish oil, lowers levels of triglycerides/cholesterol in the blood, thereby reducing/ cure heart disease, blood, joints, colon, ageing, brain and even peace of mind; and also promise in the battle against rheumatoid arthritis. the major nutritional value of linseed is given in Table 1.

Table 1: The major nutritional value of linseed

Nutritive value			**Vitamins**			**Minerals**		
Nutrients	**Quantity**	**%**	**Nutrients**	**Quantity**	**%**	**Nutrients**	**Quantity**	**%**
Total fat	42.16g	53	Thiamine	1.64 mg	38	Copper	1.12 mg	27
Dietry fibre	27.3 g	21	Vitamin E	19.95 mg	37	Manganese	2.48 mg	23
Crabohydrates	28.8 g	7	Pyridoxine	0.473 mg	10	Magnesium	392 mg	21
Protein	18.3 g	10	Folates	87 mcg	6	Iron	5.73 mg	16
Energy	534 Kcal	9	Niacin	3.08 mg	5	Zinc	4.34 mg	8
Cholesterol	0 mg	-	Riboflavin	0.161 mg	3	Calcium	255 mg	5
Total		100 %	Vitamin K	4.3 mcg	1	Total		100%
Electrolytes	Potassium	813 mg	Vitamin C	0.6 mcg	0	Omega 3 fatty acid as alpha linolenic acid		22.5 %
	Sodium	11	**Vitamin A**	0 IU	-			
			Total		100 %			

Recent advance in medical research have shown linseed as best herbal source of omega 3 and omega 6 fatty acid which have immense nutritional/medicinal effect on human body system. It has been classified as functional food because it provides numerous health benefits in addition to serving as a source of nutrients. No doubt, linseed is beneficial in several ways.

Linseed is a good source of complete protein, high order linolenic acid, complex carbohydrates, vitamins and minerals. The increasing popularity of linseed as a functional food is not only due to its high content of alfa linolenic acid (ALA) but also due to the presence of some soluble and insoluble fibre, orbitides and lignans and its suitability for use in bread, breakfast cereals and other food products. Therefore, flaxseeds are promising nutraceutical and a repository of medicinal benefits. The major health benefits of linseed are given in Table 2.

Table 2: Health benefits of linseed

Health benefits	Referance
Prevent and treatment of disorder related to immune system	Morris, 2005
Treatment of the deficiency symptoms of omega 3 fatty acid such as neurological and visual disturbance, hemorrhagic dermatitis, folliculitis and growth retardation	Georgieff and Innis, 2005; Innis, 2008
Lowering risk of cardiovascular disease, reduce cholesterol and blood sugar level, beneficial for arthritis, eczema and brain development etc.	Stahl, 2007
Reduce the risk of lupus, nephritis, arteriosclerosis and hormone dependent types of cancer	Rubilar et al., 2010
Lowering heart disease risk, risk of breast, tumor growth endometrium and prostate cancer, inhibits the development of some types of diabetes	Stahl, 2007; Mueller et al., 2010
Anticancer effects on breast, prostate and colon cancer	Zhao et al., 2007, Jhala and Hall, 2010
Regulating menstrual cycle	Garvin,2011.
Healthy balance in body and improve acne	Uzoma, 2011.
Induction in blood clotting	Jirage, 2011
Control blood pressure and decrease the burden of hypertension	Bazzano et al., 2013
Reduction in blood pressure	Ursonio et al., 2015; Caligiuri et al., 2014

Extraction of omega 3 fatty acid from linseed

Linseed oil is commonly used in food due to high percentage of omega 3 and omega 6 fatty acids. The fatty acid compositions of the oils were determined by CHNS, GC-FID. GC/MSGC-FAME and ^{1}H NMR analyzer. Among extraction methods, supercritical CO_2 is an environmentally friendly process for selectively extracting the omega 3 and omega 6 fatty acids. The extraction of linseed oil was carried out by using green solvent *viz.* supercritical CO_2 fluid, soxhlet, mechanical screw press method, conventional and accelerated method. Linseed oil is characterized by its low level of saturated and high level of polyunsaturated fatty acids. The extracted oil is superior in quality due to higher instauration/saturation ratio without solvent residue, so the oil is

most preferably used in food products.

Table 1: List of recommended linseed varieties with their salient features

Sl no.	Name of variety	Year of release & notification no.	Pedigree	Originating centre	Dura-tion	Avg. Yield kg/ha	Oil content (%)	Recommended states	Special features
(1974-1979) Vth Plan									
1	K-2	1975 44 E, 21.08.1975	Rust Resistant strain from UP x Kangra local	Ludhiana	170-175	1110(I)	40.04	Punjab, Haryana & U.P	White flower, Medium dark brown resistant to wilt and rust
2	LC-185	1975 44E, 21.08.1975	NP(RR)-37x Kangra local	Ludhiana	150-160	500(U)	38.89	Punjab	Blue flower yellow small seeded resistant to powdery mildew and rust
3	HIRA	1978 429 E, 19.12.1978	H 342 x NP(RR)-9	Kanpur	130-135	1200(R)	36.36	Bundelkhand area & U.P.	White flower, Medium dark brown seeded resistant to wilt and rust
4	MUKTA	1978 429 E, 19.12.1978	H 626 x NP(RR)-9	Kanpur	127-132	1200(I)	41.40	Eastern U.P.	Large white flower, Medium brown seeded resistant to wilt and rust

5	CHAMBAL	1978 13 E, 19.12.1978	Local x RR45	Kota	125-130	900(I)	40.11	Rajasthan	Blue flower, Brown bold seeded, moderately resistant to rust, wilt & powdery mildew
6	NEELUM	1978 13 E, 19.12.1978	T-1 x NP(RR)-9	Kanpur	140-145	1500(I)	43.00	Mid Central & Western U.P.	Blue flower, Brown bold seeded, tolerant to milt and rust
(1980-1985) VIth Plan									
7	JAWAHAR-1	1982 19 E, 14.01.1982	Selection of KP 29	Jabalpur	116-120	900(R)	38.34	Madhya Pradesh	Red violet flower, Dark brown bold seeded resistant to rust
8	JAWAHAR-7 (JLS-1)	1982 19 E, 14.01.1982	Selection of No. 55	Raipur	116-120	700(R), 300(U)	37.79	Madhya Pradesh	Red violet flower, Dark brown bold seeded resistant to rust
9	JAWAHAR-17 (JLS-7)	1982 19 E, 14.01.1982	Selection of No. 55	Raipur	117-120	1300(I), 800(R)	37.61	Madhya Pradesh	Red violet flower, Dark brown bold seeded resistant to rust
10	NEELA	1982 19 E, 14.01.1982	SPS from indigenous collection	Behrampore	127-135	850(R)	34.93	West Bengal	Blue flower, Brown medium seeded resistant to bud fly

11	LC-54	1982 19 E, 14.01. 1982	K2 x Kangra local	Ludhiana	155-165	1330(I)	42.00	Punjab, H.P. & Haryana	White flower, Brown medium seeded resistant to wilt and rust
12	C-429	1983 429 E, 03.01. 1983	No.3 x IP-135	Nagpur	125-130	1000(R)	39.07	Maharashtra	Red violet flower, Dark brown small seeded resistant to rust
13	T-397	1984 596 E, 13.08. 1984	T-491 x T-1103-1	Kanpur	120-125	1100(I)	44.00	Bundelkh and of U.P., Bihar, Assam, M.P. & Rajasthan	Violet flower, Dark brown seeded, tolerant to rust, wilt & drought
14	JAWAHAR-552(R-552)	19845 96 E, 13.08. 1984	Sel. Number 55x B-67	Raipur	118-125	900(R)	44.00	Madhya Pradesh	Violet flower, Light brown seeded, medium to rust, wilt & powdery mildew

15	PUSA-2	1985295 (E), 09.04.1985	Selection of BS 12	Kanpur	125-150	730(R)	38.31	Punjab, H.P., Haryana & Rajasthan	White flower, brown medium height seeded resistant to rust
16	PUSA-3	1985295 (E), 09.04.1985	K2 x T-603	Kanpur	125-150	800(I)	37.65	Punjab, H.P., Haryana & Rajasthan	White flower, brown medium height seeded resistant to rust
17	S-36	-	Selection of Local Solapur	Pune	130-135	400(R)	34.80	Karnataka	Blue flower, Light brown small seeded
18	HIMALINI	1985295 (E), 09.04.1985	K2 x Kangra local	Palampur	150-175	1310(I)	42.00	Punjab, H.P., Haryana & adjoining of Rajasthan	White flower, brown medium height seeded resistant to rust and wilt
19	JAWAHAR-23	1985540 (E),24.07.1985	EC 9832 x Hira	Jabalpur	120-130	1000(I)	43.00	Madhya Pradesh, Odisha, Bundelkhand of U.P., Maharashtra, Karnataka & Raj.	White flower, Brown medium seeded resistant to wilt and rust & powdery mildew
20	GARIMA	1985540 E, 24.07.1985	T-126 x Neelum	Kanpur	120-130	1490(I)	42.00	U.P.(Excl. Bundelkhand), Bihar, West Bengal & Assam	Blue flower height, brown medium tolerant to PM, seed Altenararia blight, & wilt resistant to rust
21	SWETA	1985540 E, 24.07.1985	Mukta x T-1206	Kanpur	130-135	880(R)	44.00	U.P.(Excl. Bundelkhand), Bihar, West Bengal & Assam	Blue flower height, brown medium tolerant to PM, seed Altenararia blight, & wilt resistant to rust

22	SHUBHRA	1985 540 E, 24.07.1985	Mukta x K 2	Kanpur	130-135	1390 (I), 870 (R)	45.00	U.P.(Excl. Bundelkhand), Bihar, West Bengal & Assam	Dark brown medium white flower seeded resistant to rust

(1985-1990) VIIth Plan									
23	LAXMI-27	1987 165 E, 06.03. 1987	(Neelum/R1// Neelum x NPRR-9//Neelum x R1//Neelum x Afg-8	Mauranipur	110-120	1260(I), 1020(R)	45.00	Bundelkhand of U.P.	Red violet flower, dark brown medium height seeded resistant to rust, wilt & PM
24	GAURAV	1987 834 E, 18.09. 1987	Sel.3 x EC-1552	Kanpur	137-140	1050(seed) 950(fibre)	43.00	U.P.(Excl. Bundelkhand), Bihar, West Bengal & Assam	Violet red flower, brown medium seeded resistant to wilt and rust
25	KIRAN	1988 10 E 01.01. 1988	(R1x Afg-8) x R1	Raipur	120-126	750(R)	43.00	Madhya Pradesh, Odisha, Bundelkhand of U.P., Maharashtra, Karnataka & Raj.	Violet red flower, brown medium seeded resistant to wilt , powdery mildew and rust
26	JANKI	1988	New River x LC 216	Palampur	165-170	1200(I)	42.00	Himachal Pradesh (Kanger valley)	Violet red flower, brown medium seeded resistant to wilt , powdery mildew and rust
27	JEEVAN	1988 10 E 01.01. 1988	Summit x LC-216	Palampur	175-180	1090(seed) 1100(fibre)	45.00	Punjab and Himachal Pradesh	Violet red flower, brown medium seeded resistant to wilt , powdery mildew and rust
(1992-1997) VIIIth Plan									

28	SURABHI	1995 408(E), 04.05. 1995	LC 216 x LC 185	Palampur	165-170	1000(U)	44.00	Himachal Pradesh	White flower, yellow small seeded resistant to wilt, powdery mildew and rust
29	NAGARKOT	1995 400 E, 04.05. 1995	New river x LC 216	Palampur	165-170	1150 (Seed) 950 (Fibre)	43.00	Panjab, U.P., H.P., W.B., Assam, Haryana and Rajasthan (Kota)	Blue flower, brown medium seeded resistant to wilt powdery mildew and rust
30	SHIKHA	1997 401 E, 15.05. 1998	Hira xCrista	Kanpur	135-140	1233 (Seed) 1033 (Fibre)	42.00	U.P.(Excl. Bundelkhand), Bihar, W.B. and Assam	Blue flower, brown medium seeded resistant to wilt and rust
(1997-2002) IX th Plan									
31	LC 2023	1998	Flake x LC 54	Ludhiana	158-163	1294 (I)	38.8	Punjab	
32	PADMINI	1998 1050 (E) 26.10. 1999	(EC 41628 x EC 77959) X (DPL 20 x Neelum)	Manranipur	120-125	943 (R)	43.00	M.P., Rajasthan, Bundelkhand of U.P., M.H., Karnataka and Odissa	Blue flower, brown seeded resistant to powdery mildew
33	JLS- 9	1998 425 E, 08.06. 1999	(RL -102 x R7) x J 23	MP	115-125	1250 (I), 1000 (R)	42.00	M.P.	White flower height brown and medium bold seeded resistant to wilt, Powdery mildew and rust

34	RASHMI	1999 1050 (E), 26.10. 1999	Gaurav x Janki	Kanpur	135-140	1003 (seed), 719 (fibre)	41.00	U.P., Bihar, W.B. and Rajasthan (Kota)	Very light blue flower, brown seeded resistant to wilt, Powdery mildew and rust
35	MEERA	2000 340 E, 03.04. 2000	(RL75-6-2 x RL 29-8) x LCK 8528	Kota	135-140	1439 (Seed), 1011 (fibre)	42.00	U.P., Bihar, W.B. and Rajasthan (Kota)	Violet blue flower, brown seeded resistant to wilt, Powdery mildew and rust
36	RL-914	2000 973 E, 04.09. 2002	(RR 9 x R-93) x (Flake 1 x LC 54)	Kota	135-140	1617(I)	41.10	Kota command area of Rajasthan	Violet blue flower, height brown seeded resistant to wilt, Powdery mildew and rust
37	PARVATI	2001 92 E, 02.02. 2001	EC 41628/EC77959//DPL20 x Neelam///EC216 x Hira// BR 1x NP 440	Manranipur	140-146	1600 (seed), 1020 (fibre)	42.00	U.P., Bihar, W.B. and Rajasthan (Kota)	Blue flower, height brown mediun seeded resistant to Powdery mildew and rust
38	SHEELA	2001 92 E, 02.02. 2001	Gaurav x Janki	Kanpur	155-160	1379 (R)	41.00	Himachal Pradesh, Panjab, Haryana and J& K	Blue flower, Shining brow seeded resistant to wilt, powdery mildew and rust
39	SHEKHAR	2001 92 E, 02.02. 2001	Laxmi 27 x EC 1387	Kanpur	135-140	1555(I), 920 (R)	43.00	U.P.(Excl. Bundelkhand), Bihar, W.B. and Assam	Violet blue flower, Shining moderately resistant brown seeded to bud fly,

									Alternaria blight & Powdery mildew and rust
40	NL-97	2001 92 E, 02.02. 2001	R-7 x RLC 4	Nagpur	115-120	641 (R)	42.00	Bidarbha Region	Blue flower, height brown mediun seeded
(2002-2007) Xth Plan									
41	SUYOG (SLS -27)	2002 161 E, 04.02. 2004	(Kran x KL-168) x Kiran	Sagar	118-125	1509 (I)	41.43	Rajasthan, BKD region of U.P., M.P., Maharastra, C.G., Odissa, A.P., & Karnataka	White flower, light brown seeded, resistant to rust, powdery mildew and bud fly
42	BINWA (KL-210)	2004 122 E, 02.02. 2005	Flake 1 x SPS-47/7-10-3	Palampur	179-186	858 (I)	40.00	Haryana, Panjab, Himachal Pradesh and J&K	Blue flower, yellow seeded resistant to wilt and rust
43	BANER (KL-224)	2005 1177 (E), 25.08. 2005	EC 21741 x LC 216	Palampur	171-203	511 (U)	39.70	Haryana, Panjab, Himachal Pradesh and J&K	Purple flower, brown coloured, seed resistant to rust

44	INDIRA ALSI- 32 (RLC-76)	2005 1566 (E), 05.11. 2005	Kiran x RLC 29	Raipur	110-115	780(R)	39.18	CG, Maharastra, Karanatka & Odisha	Blue flower, dark brown seeded, powdery mildew
45	KARTIKA (RLC-76)	2005 1566 (E), 05.11. 2005	Kiran x LCK-88062	Raipur	103-108	1078(R)	42.93	Rainfed areas of Chhattisgarh	Blue flower, light brown medium seed and bud fly resistant to Powdery mildew
46	DEEPIKA (RLC 78)	2006 1178 (E), 20.07. 2007	Kiran x Ayogi	Raipur	110-115	1272 (SI&U)	41.39	Partially irrigated as well as *Utera* Situation of CG	Blue flower, brown seeded resistant to powdery mildew
47	SHARDA (LMS-4-27)	2006 1572 (E), 20.09. 2006	(Shubhra x J1) x (J1 x Kiran)	Manranipur	100-105	762 (R)	41.32	CG, Maharashtra, Karnataka, AP & Odissa	White flower brown seeded MR to wilt, Powdery mildew and bud fly
(2007-2012) XI th Plan									
48	PRATAP ALSI-1 (RLU-6)	2007 1703 (E), 05.10. 2007	Acc. 750 x RL 29-8	Kota	129-135	1997 (seed) 834 (fibre)	41.08	Rajasthan Kota command area	White flower, brown seeded MR to rust, wilt and bud fly

49	LC-2063	2007 1108 (E), 08.05. 2008	1509 x LC 54	Ludhiana	115-125	1200 (I)	37.40	Irrigated area of Punjab State	Blue flower, Dark Brown seeded MR to bud fly
50	HIMANI KL- 214	2008 2458 (E), 16.10. 2008	DPL 20 x KLS-1	Palampur	177-200	583 (*U*)	36.40	HP, PB, Haryana and J&K	Blue flower, small Brown seeded MR to Powdery mildew and rust
51	AZAD ALSI-1 (LMS 9-2K)	2008 2458 (E), 16.10. 2008	RL 904 x Kiran	Manranipur	125-130	1610 (I)	39.92	BKD area of UP, MP and Rajasthan	Red violet MR to wilt, PM blue flower, dark brown seeded resistant to bud fly & rust
52	RLC 92	2008 2458 (E), 16.10. 2008	Jeevan x LCK 9209	Raipur	111	1196 (I)	37.70	CG, Maharashtra, Karnataka, AP & Odissa	Ting blue flower, light violet flower, brown medium seeded MR to Powdery mildew rust and bud fly
53	SHIVAL (SLS- 67)	2010 733 (E), 01.04. 2010	LCK 9610 x LMS 127	Sagar	108-110	1252 (R)	40.16	BKD area of UP, MP and Rajasthan	White flower, light brown seeded MR to powdery mildew and rust
54	JAWAHAR LINSEED -41	2011 2326(E), 10.10.	Kiran x Acc. No.443	Hoshangabad	115-120	1600 (I)	40.00	Area of MP state with limited facility	White flower, brown bold seeded resistant to rust, powdery mildew and rust

	(PKDL-41)	2011							
55	BHAGSU (KL-215)	2010 2137 (E), 31.08. 2010	RL-50-3 x Surbhi	Palampur	175-201	428(*U*)	36.38	Himachal Pradesh, J&K, Uttaranchal, Punjab and Haryana	MR to rust, blue flower, brown small seed, MR to rust
56	RUCHI (LCK 5021)	2011 283 (E), 07.02. 2011	Garima x LCK 88062	Kanpur	134	1366 (seed), 1055 (fibre)	39.84	UP.(Excl. Bundelkhand), Bihar, WB and Assam	Blue flower, seed shining brown, MR to rust powdery mildew and bud fly

57	SLS 73	2011 632(E) 25.03. 2011	Padmini x Laxmi 27	Sagar	111-114	1090 (R)	38.82	BKD area of UP, MP, Rajasthan	Large blue flower, light brown seeded resistant to rust, powdery mildew and bud fly
58	MAU AZAD ALSI-2 (LMS 149-4)	2011 2326(E) 10.10. 2011	KL 178 X Hira	Manran ipur	105-110	815 (R)	40.17	CG, Maharashtra, Karnataka, AP, Odisha	Medium white flower, medium brown seeded resistant to rust
59	NNDL 2004-05	2011	Garima x RL 993	Faizaba ad	125-130	1800(I)	40.0	Eastern Uttar Pradesh	Light flower, Medium dark brown seeded resistant to Alternaria blight, rust and powdery mildew
60	NDL 2002	2011	Garima x EC 44	Faizaba ad	130-135	1600 (I & R)	39.3	Eastern Uttar Pradesh	Light blue flower, Medium dark brown seeded resistant to rust, powdery mildew and tolerant to bud fly
(2012-17) XII th Plan									
61	Pratap Alsi 2(RL 26016)	2012 268(E) 28.1.2 015_	RL 914 x NL 93	Kota	129-135	1957 (I)	41.81	Rajasthan	Blue flower Shining brown bold seeded, PM, wilt and bud fly. Moderate resistant to Alternaria blight, powdery mildew, wilt and bud fly
62	PKVNL-260	2015 112(E) 12.01. 2015	R 552 x RLC6	Nagpur	102-106	963 (R)	37.67	Maharashtra	Light flower, brown seeded MR to powdery mildew and bud fly

63	Kota Barani Alsi-3 (RL-292002)	2015	RL 903 x Ayogi	Kota	119-124	1370 (R)	38.73	Rajasthan	Blue flower, Dark brown shining seeded, Resistant to rust and moderately resistant to Altarnaria blight, powdery mildew and bud fly
64	Kota Barani Alsi-4 (RL-10193)	2015	Triveni x RL-1011	Kota	120-126	1100 (R)	40.37	UP, MP, Rajasthan	White flower, Shining dark brown seeded moderately resistant to Altarnaria blight and Powdery mildew
65	Chattisgadh Alsi - 1 (RLC-133)	2015	NL-14 x ACC-26	CG	110-113	844 (R)	36.4	CG	White flower, Early, Moderatety resistant to bud fly, Altarnaria blight, Powdery mildew and rust
66	Tiara (JRF-2)	2015 1228(E)	FT-889 x FT-895		135-150	1290		Himachal Pradesh. Uttarakhand. Sikkim, Arunachal Pradesh, Uttar Pradesh, Kashmir and West Bengal.	

67	Arpita (OL 98-13-1)	2016 3540(E), 22.11.2016	RLC 29 x R 1871	Keonjhar	102-106	849 (R)	35.67	Odisha	Blue flower, light brown seeded, resistant to wilt and powdery mildew
68	Utera Alsi (RLC-143)	2016	LCK-88062 x T-397	CG	115-118	570 (N)	34.1	CG, MP, UP, JH, Maharastra and Odisha	Suitable of utera para situaton, Blue flower, resistant to powdery mildew and rust, moderately resistant to budfly and Altarnaria blight
69	JLS-79	2016 3540(E), 22.11.2016	Padmini x Laxmi-27	Sagar		1762	40.35 %	Madhya Pradesh	
70	PKV-NL-260 (NL-260)	2016 3540(E), 22.11.2016	R552 x RLC-6	Nagpur	102-106		35.56 %	Maharashtra	
71	Divya (BAU-06-03)	2016 3540(E), 22.11.2016	BAU-1008 x Kiran	Kanke	180	1038	41.3%	Punjab, Haryana, Himachal Pradesh and Uttarakhand.	
72	Indu (LCK	2016 1007(	LMS1-2K**x** LMS4-27	Kanpur	133-137	955 (I)	40.00	Whole of U.P.	Blue flower, bold brown seeded,

	1108)	E) 30.03.17							highly resistant to rust, resistant to A.B. & P.M.
73	UMA (LCK-1101)	2016 1007(E) 30.3.17	Polf 34 **x** Padmini	Kanpur	120-123	868 (R)	37.6	Whole of U.P.	Blue flower, medium light brown seeded, moderately resistant to rust, wilt & A.B.
74	Varsha Alsi (RLC-148)	2017	SIKO 10 x Kiran	CG	110-114	1033.5 (R)	35.7	CG, MP, UP, JH, Maharastra and Odisha	Suitable for rainfed situation moderately resistant to budfly, moderately susceptible to wilt and powdery mildew
75.	Priyam (BAU-2012-1)	2017 2805(E), 25.08.2017	BAU-2k-14 x Garima	Ranchi (Jharkhand)	173	1151	37.15 %	Punjab, Haryana, Himachal Pradesh and Uttarakhand.	
76.	Jawahar Linseed Sagar - 95 (JLS -95) (SLS- 95)	2018 1379(E), 27.03.2018	JLS-27x GS-281	Sagar	133	1240	39.4%	Bundelkhand part of Uttar Pradesh, Rajasthan Madhya Pradesh and Central Peninsular India (Zone-lll).	

77	JLS-66 (SLS 66)	2018 399(E), 24.01. 2018		Sagar	114	2200	40.5%	Madhya Pradesh.	
78	Sabour Tisi-1. (BAUP-101)	2018 1379(E), 27.03. 2018	Selection from Kishanganj local	Patna	124	453	32.68 %	Uttar Pradesh excluding Bundelkhand, Bihar, Jharkhand, West Bengal and Assam.	
79	Varsha Alsi (RLC-148)	2018 1379(E), 27.03. 2018	SJKO-10 x Kiran	Raipur		1237	36.81 %	Rajasthan, Madhya Pradesh, Bundelkhand region of U.P., Odisha, Maharashtra, Chhattisgarh and Karnataka.	
80	Him Palam Alsi-2 (KL-263)	2018 1379(E), 27.03. 2018	KL-223 x KL-224	Palampur		1633	35.58 %	Himachal Pradesh, Haryana and Jammu and Kashmir.	
81	RAJAN	2018 1498(	Gaurav **x**GS 234	Kanpur	130-135	1528(S) 1093(F)	38.01	Whole of U.P.	Blue flower, brown medium seeded,

		E) 1.4.19							highly resistant to rust, moderately resistant to A.B.
82	SURYA	2018 3220(E) 5.9.19	Shubhra **x** H-25	Kanpur	155-160	1431(I)	35.49	H.P., Punjab, Haryana and J&K	Blue flower, brown medium seeded, highly resistant to rust, moderately resistant to wilt
83	Utera Alsi-2 (RLC-153)	2019 The Gazett No. 1326; 02.04. 2019	LCK 88062 x EC-1424	Raipur	118-120	519	34.8	Assam, Bihar, Chhattisgarh, Jharkhand, Orrisa, Madhya Pradesh, Maharashtra & Uttar Pradesh	
84	LSL-93	2019 3220(E), 05.9.2019	Selection from SLS-93	Latur	106	846	37.07 %	Maharashtra.	

Table 2: Agronomical Practices for linseed in different crop growing situations

State	Situation	Varieties	Optimum time sowing	Row Spacing (cm)	Seed rate (kg/ha)	Fertilizer N:P (kg/ha)
Assam	Rainfed	Shekhar, Sweta, Shubhra, T397, Sabour Tisi-1, Utera Alsi	Ist fortnight of October	25	30	40:20
	Irrigated	Shekhar, Garima, Shubhra,	IIndfortnight of October	25	25-30	60-80:30
	DP	Ruchi, Rashmi, Meera, Shikha, Gourav, Parvati	Ist week of November	20	45	80:30
Bihar	Rainfed	Shekhar, Sweta, Shubhra, T397, Sabour Tisi-1(BAUP-101)	Mid October	25	30	40:20
	Irrigated	Shekhar, Garima, Shubhra,	IInd fortnight of October	25	25-30	60-80:30
	DP	Ruchi, Rashmi, Meera, Shikha, Gourav, Parvati	IInd fortnight of October	20	45	80:30
Chhattisgarh	Rainfed	Indira Alsi-32, Kartika, Man Azad Alsi-2, Padmini, Kiran, T397, Chattisgarh Alsi-1(RLC-133), Varsha Alsi(RLC-148)	IInd fortnight of October	25	25-30	40:20:20
	Utera	R552, Deepika, RLC-143, RLC-153(Utera Alsi-2)	IInd fortnight of October	Broadcast	35-40	20:0:0
	Irrigated	RLC-92, Suyog, JLS-9, J23 and T397	Mid November	25	25-30	60:30:30
Haryana	Irrigated	Binwa, Himalini,LC-54, Surya	IInd fortnight of October to Ist week of November	25	20-25	60-80:30
	DP	Nagarkot, Jeevan	IInd fortnight of October to Ist week of November	20	45	80:30
H.P. and J&K	Rainfed	Surbhi, Janki, Sheela, Priyam(BAUP-2012-01)	IInd fortnight of October	25	25-30	40:20

	Utera	Bhagsu, Himani, Baner, Surabhi, Himalini, RLC-143	October	Broad cast	50-60	20:00
	Irrigated	Binwa, Janki, Himalini,LC-54, Surya	IInd fortnight of October to Ist week of November	25	20-25	60-80:30
	DP	Nagarkot, Jeevan, (JRF-2)	Mid October	20	65-75	90:30
Jharkhand	Rainfed	Shekhar, Sweta, Shubhra, T397, RLC-148	Mid October	25	30	40:20
	Irrigated	Shekhar, Garima, Shubhra, T397	Ist fortnight of October	25	25-30	60-80:30
	DP	Ruchi, Rashmi, Meera, Shikha, Gourav, Parvati	Ist fortnight of October	20	45	80:30
Karnataka	Rainfed	NL-115, Indira Alsi-32, Man Azad Alsi-2, Padmini, Kiran, J-23, Sharda, RLC-148	Up to Ist week October	25	25-30	40:20
	Irrigated	NL-115, RLC-92, Suyog, J-23	Up to Mid October	25	25-30	60-80:30
MP	Rainfed	SLS-73, Padmini, Kiran, T-397, SLS-67, JLS-95, JLS-66, RLC-148, RL-10193	Ist fortnight of October	25	25-30	30:15
	Utera	R-552, RLC-153	Ist to 3rd of October	Broad cast	25-30	10-20:00
	Irrigated	PKDL-41, Man Azad Alsi-2, Suyog, JLS-9, J-23, T-397, JLS-66, RL-10193	Mid October	25	25-30	60-80:30
Maharashtra	Rainfed	Man Azad Alsi-2, Indira Alsi-32, Padmini, Kiran, J-32, PKV-NL-260, Sharda, RLC-148	Ist fortnight of October	25	30	40:20
	Utera	R-552, PKV-NL-260	Ist fortnight of October	Broad cast	35-40	10-20:00
	Irrigated	RLC-92, Suyog, J-23	2nd to 3rd week of October	25	25-30	60-80:30
Odisha	Rainfed	Man Azad Alsi-2, Indira Alsi-32, Padmini, Kiran, J-32, Sharda,	Ist fortnight of October	25	25-30	30:15

		Arpita, RLC-143				
	Utera	RLC-148, RLC-153	Ist fortnight of October	Broad cast	35-40	10-20:00
	Irrigated	RLC-92, Suyog, J-23, Arpita	Mid October	25	25-30	60-80:30
Punjab	Irrigated	Binwa, LC-54, Himalini, LC2063, Surya(LCK-1404), Divya, Priyam	Mid October to Mid November	25	25-30	60-80:30
	DP	Nagarkot, Jeevan	Last week of October	20	50-60	80:30
Rajasthan	Rainfed	SLS-73, RL914, Padmini, Kiran, SLS-67, SLS-95, RLC-148, Kota Barani Alsi-4, Kota Barani Alsi-3	Mid October	25	25-30	40:20
	Irrigated	Suyog, J-23	2nd fortnight of October	25	25-30	60-80:30
	DP	Meera, Rashmi, Shikha, Pratap Alsi-1	2nd fortnight of October	20	45	80:30
UP(Excluding Bundelkhand)	Rainfed	Shekhar, Sweta, Shubhra, Uma	Mid October	25	25-30	40:20
	Irrigated	Shekhar, Garima, Neelum, Indu	2nd fortnight of October	25	25-30	60-80:30
	DP	Rashmi, Parvati, Shikha, Gaurav, Ruchi, Rajan	2nd fortnight of October	20	45	80:30
Bundelkhand of UP	Rainfed	Shekhar, Sweta, Shubhra, Uma	1st fortnight of October	25-30	25-30	40:20
	Irrigated	Shekhar, Garima, Neelum, Indu	Mid October	25-30	25-30	60-80:30
	DP	Rashmi, Parvati, Shikha, Gaurav, Ruchi, Rajan		20	45-50	80:30
West Bengal	Rainfed	Shekhar, Shubhra, Sweta, Sabour Tiai-1(BAUP-2012-1)	1st fortnight of October	25	25-30	40:20
	Irrigated	Shekhar, Shubhra, Garima	2nd fortnight of October to 1st fortnight of November	20	25-30	60-80:30
	DP	Ruche, Rashmi, Meera, Shikha, Gaurav, Nagarkot, Parvati, Tiara(JRF-2)	2nd fortnight of October to 1st fortnight of November	20	45	80:30

Table 3: Insect-pests of linseed and systematic approach for their management

Time of practices/ growth stage	Target Pests	Control measures
Pre-sowing	Bud fly and other insect-pests	Soil solarization; Avoid continuous cultivation of linseed in the same field.
Land preparation	Termite	Apply methyl chlorpyriphos dust 1.5% @ 25-30kg/ha in soil, when termites are regular and heavily damaging pest.
Sowing	All major pests	Sowing may be restored 10-15 days earlier to minimize the bud fly infestation; Selection of suitable resistant varieties like Neela, Kiran and Jawahar-1 or moderately resistant high yielding varieties like Shekhar, parvati, Sharada, PKVNL-260 and Kartika.
Seedling stage	Cut worm and Termite(Moderate infestation)	Dusting of the crop with methyl parathion 2% @ 25kg/ha.
Vegetative stage	Leaf miner, Sap sucking pests and Defoliators(Moderate infestation)	Apply Dimethoate 30EC (0.03%) or monocrotophos 36EC (0.04%) as and when required.
Flowering stage	Bud fly, Leaf miner, Defoliators and Sap sucking pests(6.0% ETT, for bud fly and moderate to severe damage by Budfly & other pests)	The economic threshold level of 8.77 per cent bud fly infestation is recommended for the management of linseed bud fly. Use of light trap for bud fly; Use of attractant (1 kg jaggery in 75 lit. of water) for bud fly; Use of bamboo/wodden pegs as dead perches for predator birds; spray fortnightly neem based formulation (0.5%) or spray Imidacloprid 17.8 SL (0.004%) or Spinosad 45 SC (0.015%) alone or with Mancozeb (0.2%); Control leaf miner, defoliator and other sap sucking pests as per above given schedules.

Bud initiation	Bud fly	• Two fortnightly sprays of Azadirachtin 300 ppm @ 5 ml/lit or Acetamiprid 20 SP @ 0.004% (0.2 g/lit) or Imidacloprid 17.8 SL @ 0.0045% (0.2 ml/lit) starting from bud initiation is recommended for the effective management of linseed bud fly. • Cultivation of improved variety of linseed at normal sowing time in irrigated and rain fed situation along with recommended agronomic practices and two alternate fortnightly spray of insecticides (imidacloprid 17.8 SL @ 0.004%) and botanical (nimbicidine 300 ppm (0.5%) or vice-versa reduced 64-70% bud fly infestation with 148-203% higher seed yield in irrigated and 54- 68% lower bud fly infestation with 79- 491% more yield in rain fed situation.
Capsule formation stage	Gram pod borer (Moderate to heavy infestation)	Apply HaNPV @ 250-300 LE/ha.

Table 4: Diseases of linseed and Systematic approach for their management

Disease with causal organism	**Prevalence in the country/specific area**	**Resistant/ tolerant varieties**	**Control measures**
Wilt (*Fusarium oxysporum*)	Incidence throughout the country but more prevalent in central part. Average loss 12-15% if continues certification loss upto 70%	Himalini, LC-54, Sheela, Nagarkot, Kiran, Surabhi	• Soil solarization. • Avoid continuous cultivation of linseed in the same field. • 2-3 years crop rotation is most effective prevention. • Use of the resistant varieties.

			• Seed treatment with Thiram (3g/kg seed) or *Trichoderma viride* or *Trichoderma harzianum* 4g/kg seed. • Use of well rotted FYM treated with *Trichoderma* then, properly mixing with soil at the time of last ploughing.
Alternaria blight(*Alternaria lini and Alternaria linicoa*)	Prevalent in Northern region having humidity range of 90-95% and temperature 25-30°C. average loss 20-35% if atmospheric conditions suitable loss upto 60%	Parvati	• Use of resistant varieties, Like Jeevan, Gaurav. • Seed treatment with Thiram 3g/kg seed with followed by two spray of Mancozeb after 30 days of sowing as prophylactic spray and 2nd after symptom appearance.
Rust (*Melampsora lini*)	Prevalent in Nrthern part of the country and serious in colder hilly regions. Loss only hilly region 15-20%.	Himalini, LC-54, Sheela, Nagarkot	• Use of resistant/ tolerant cultivars. • Destroy plant debris and weeds to reduce the primary source of infection. • Use of clean seed, seed treatment with Oxycarboxin. • Spray of calixin+Dithane M-45 (0.25%) at 15 days interval control the disease.

Powdery Mildew (*Oidium lini*)	Central part of country loss were depend upon time of sowing Early sown (15 Oct.) No loss late sown(25Oct.) Loss up to 20%	Himalini, LC-54, Sheela, Nagarkot	• Use of resistant/ tolerant cultivars, early sowing. • 2-3 sprays of Calixin(0.05%) or Karathane (0.1%), Wettable sulphur(0.3%) reduces the disease. • Two sprays of Propiconazole (0.1%) or Difenconazole (0.05%) or wettablesulphur (0.25 %) or Quintal (0.1 %) or Hexaconazole (0.1%) one at disease initiation and second 15 days later.
Dodder/Cascuta (*Cascuta hylina*) (Plant parasite)	Chhattisgarh, M.P., Odisha and Vidarbha region of Maharashtra. Losses up to 5-8%		• Mechanical removing of parasitic vine from fields and parasitic seeds from seed lot. • Preventing the movement of grazing animals from infested fields. • Restricting the flow of irrigation water from infested area, Spray 2,4-D @ 0.5kg/ha.

IMPACT OF IMPROVED TECHNOLOGIES ON THE PRODUCTIVITY POTENTIAL OF LINSEED UNDERREALFARMSITUATION
STATE-WISE YIELD GAP 2013-14

State	Yield (kg/ha)		Yield Gap
	Improved Technology (IT)	Farmers' Practice (FP)	
Irrigated			
Uttar Pradesh	1390	560	59.73
Bihar	876	653	25.46
Madhya Pradesh	1575	1050	33.33
Chhattisgarh	640	350	45.33
Himachal Pradesh	1089	495	54.55
Rajasthan	1461	1224	16.25
Rainfed			
Uttar Pradesh	913	617	32.50
Bihar	876	653	25.46
Jharkhand	584	383	34.42
Madhya Pradesh	1150	735	36.09
Chhattisgarh	649	476	26.60
Himachal Pradesh	875	447	48.91
Rajasthan	1217	1023	15.94
Maharashtra	943	730	22.61
Odisha	780	555	28.85
Karnataka	589	478	18.85
Nagaland	892	740	17.04
Assam	745	509	31.68

State	Yield (kg/ha)		Yield Gap
	Improved Technology (IT)	Farmers' Practice (FP)	
Utera			
Uttar Pradesh	>94	466	41.43
Jharkhand	413	305	26.15
Chhattisgarh	390	310	20.60
Himachal Pradesh	522	235	54.98
Madhya Pradesh	400	300	25.00
Odisha	710	365	48.59

State	Area (000 ha)	Production (000tonnes)	Yield (kg/ha)	Potential Districts
Madhya Pradesh	110.4	55.0	498	Rewa, Balaghat,Sidhi, Chhatarpur, seoni, Damoh, Satna, Dindori, Shahdol, Katni
Maharashtra	31.0	8.0	258	Chandrapur, Bhandra, Nagpur, Osmanabad, Aurangabad, Bid, Parbhani, Latur
Chhattisgarh	26.2	11.1	424	Durg, Rajnandgaon, Kawardha, Dhamtari, Raipur, Janjgir-Champa, Korba
Uttar Pradesh	26.0	10.0	385	Sonbandra, Mahoba, Mirzapur, Hamirpur, Banda, Allahabad, Jhansi, Chandauli
Bihar	18.7	15.9	850	Aurangabad, Pashchim Champaran, Kaimur(Bhabua), Gaya, Rohtas, Kishanganj
Odisha	22.9	11.0	478	Mayurbhanj, Kalahandi, Nawapara, Navranpan, Keonjhar, Puri
Jharkhand	25.5	14.5	568	Paschim, Singhbhumi, Palamau, Godia
Karnataka	6.0	2.0	333	Bijapur, Bagalkot, Belgaum, Koppal, Gulberga, Gadag
Assam	6.0	3.9	643	Barpeta, Nalbari, Dhubri, Kamrup, Marigaon
Nagaland	5.8	4.6	803	Kohima, Wokha, Dimapur, Mokokchung, Zunhelxolo, Phok, Tuensang
West Bengal	10.1	2.0	200	Uttar Dinajpur, Nadia, Jalpaiguri
Rajasthan	2.1	2.9	1351	Kota, Baran, Bundi
Himachal	1.1	0.3	255	Kangra and Chamba

Pradesh				
All India	292.1	141.2	484	

STATEWISE AREA, PRODUCTION AND YIELD OF LINSEED ALONG WITH POTENTIAL DISTRICTS 2013-14

प्रदर्शन - अकोली

RLC-143

RLC-153 (*Utera Alsi 2*)